

Docker Task -3 - Sudir

Task Description:

Create a custom docker image for nginx and deploy it using docker compose, where the volume bind mount should be at /var/opt/nginx location. Push the created custom docker image to your docker-hub.

Techstacks needs to be used :

- AWS EC2
- Docker
- Docker Compose

How do I submit my work?

- Push all your work files to GitHub (Code & O/P screenshot images must).
- Submit your URLs in the portal.

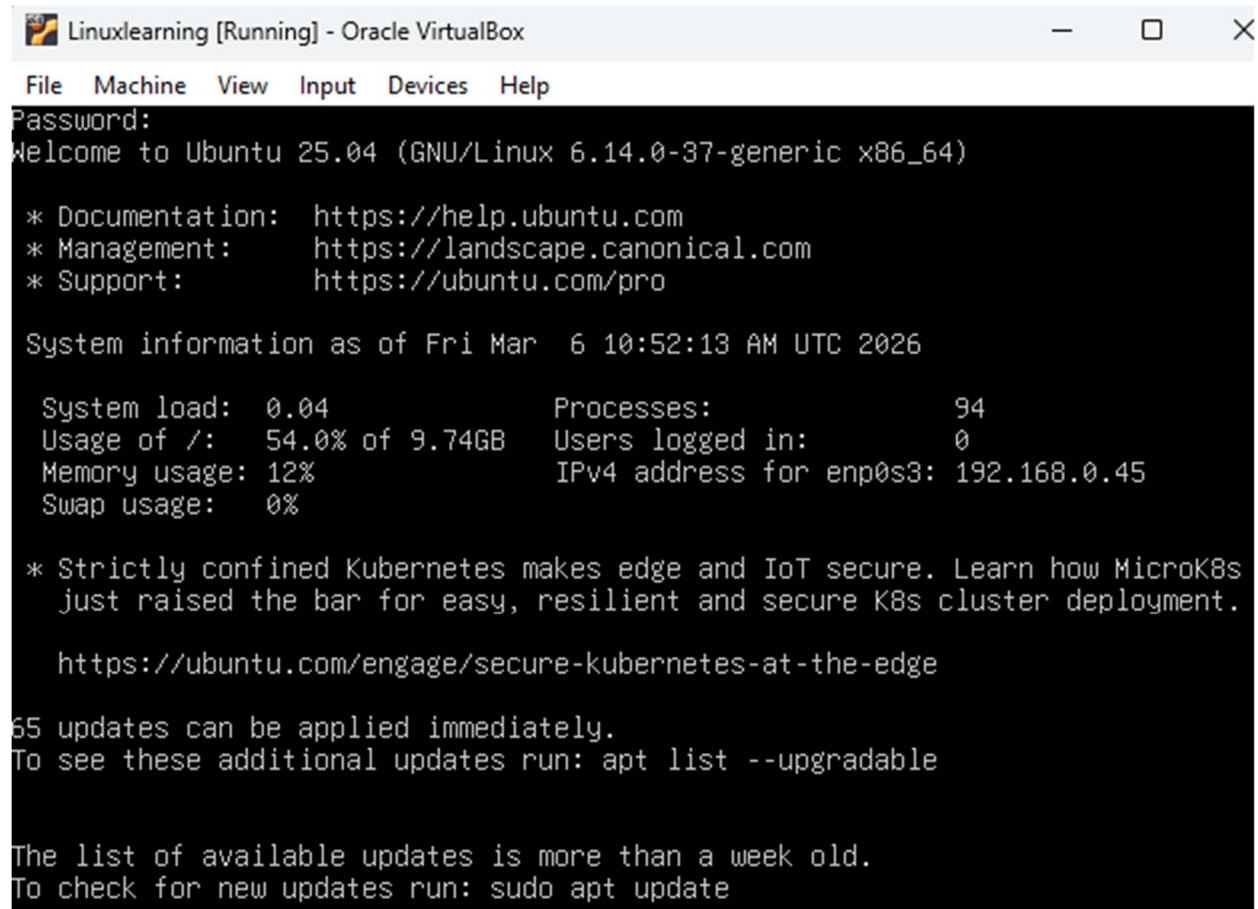
Terms and Conditions?

- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

Virtual Machine Setup

VM running in oracle virtual box



```
Linuxlearning [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Password:
Welcome to Ubuntu 25.04 (GNU/Linux 6.14.0-37-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri Mar  6 10:52:13 AM UTC 2026

System load:  0.04               Processes:           94
Usage of /:   54.0% of 9.74GB    Users logged in:    0
Memory usage: 12%               IPv4 address for enp0s3: 192.168.0.45
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

65 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
```

```
root@learninglinux:/home/docker-nginx-task# hostname
learninglinux
root@learninglinux:/home/docker-nginx-task# whoami
root
root@learninglinux:/home/docker-nginx-task# exit
exit
root@learninglinux:~# whoami
root
root@learninglinux:~# exit
logout
sudir@learninglinux:~$ sudir
```

MY IP (bridged adapter in network settings)

```
sudir@learninglinux:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:f5:cb:2b brd ff:ff:ff:ff:ff:ff
    altname enx080027f5cb2b
    inet 192.168.0.45/22 metric 100 brd 192.168.3.255 scope global dynamic enp0s3
        valid_lft 84981sec preferred_lft 84981sec
```

Install Docker

Update package repository.

sudo apt update

Install Docker.

sudo apt install docker.io -y

Start Docker service.

sudo systemctl start docker

sudo systemctl enable docker

Verify installation.

```
sudir@learninglinux:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~25.04.1
sudir@learninglinux:~$
```

Install Docker Compose

sudo apt install docker-compose -y

Verify installation.

```
sudir@learninglinux:~$ docker-  
docker-compose version 1.29.2,
```

```
sudir@learninglinux:/home/docker-nginx-task$ ll  
total 28  
drwxr-xr-x 5 root docker 4096 Mar  6 10:59 ./  
drwxr-xr-x 5 root root  4096 Mar  6 10:56 ../  
-rw-r--r-- 1 root docker  163 Mar  6 10:59 docker-compose.yml  
-rw-r--r-- 1 root docker  186 Mar  6 10:58 Dockerfile  
drwxr-xr-x 2 root docker 4096 Mar  6 10:57 html/  
drwxr-xr-x 2 root docker 4096 Mar  6 10:58 nginx/
```

docker-coposer.yml

```
sudir@learninglinux:/home/docker-nginx-task$ cat docker-compose.  
version: "3"  
  
services:  
  
  nginx-server:  
  
    build: .  
  
    container_name: custom-nginx  
  
    ports:  
      - "8080:80"  
  
    volumes:  
      - ./html:/var/opt/nginx
```

Dockerfile

```
sudir@learninglinux:/home/docker-nginx-task$ cat Dockerfile  
FROM nginx:latest  
  
RUN mkdir -p /var/opt/nginx  
  
COPY nginx/nginx.conf /etc/nginx/nginx.conf  
COPY html/index.html /var/opt/nginx/index.html  
  
EXPOSE 80  
  
CMD ["nginx", "-g", "daemon off;"]
```

UI

```
sudir@learninglinux:/home/docker-nginx-task$ cat html/index.html
<html>
<head>
<title>Nginx Docker</title>
</head>

<body>
<h1>Nginx running from custom docker image</h1>
<h2>Bind mount location: /var/opt/nginx</h2>
</body>

</html>
```

Nginx conf

```
sudir@learninglinux:/home/docker-nginx-task$ cat nginx/nginx.conf
events {}

http {

    server {

        listen 80;

        location / {
            root /var/opt/nginx;
            index index.html;
        }

    }

}
```

```
root@learninglinux:/home/docker-nginx-task# docker-compose build
Building nginx-server
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  6.656kB
Step 1/6 : FROM nginx:latest
latest: Pulling from library/nginx
206356c42440: Pulling fs layer
b47f187216b6: Pulling fs layer
1ad233904a11: Pulling fs layer
eedda9fd8786: Pulling fs layer
35ff83c394d6: Pulling fs layer
17d0911eaf62: Pulling fs layer
```

```
Status: Downloaded newer image for nginx:latest
---> fd204fe2f750
Step 2/6 : RUN mkdir -p /var/opt/nginx
---> Running in 3094748a3fd5
---> Removed intermediate container 3094748a3fd5
---> 37a3a094ebe7
Step 3/6 : COPY nginx/nginx.conf /etc/nginx/nginx.conf
---> 518c580a1106
Step 4/6 : COPY html/index.html /var/opt/nginx/index.html
---> 99054a516113
Step 5/6 : EXPOSE 80
---> Running in 6084005f5bel
---> Removed intermediate container 6084005f5bel
---> a88dlafa6a94
Step 6/6 : CMD ["nginx", "-g", "daemon off;"]
---> Running in 22208501e347
---> Removed intermediate container 22208501e347
---> 3e5122f0cc6f
Successfully built 3e5122f0cc6f
Successfully tagged docker-nginx-task_nginx-server:latest
```

```
root@learninglinux:/home/docker-nginx-task# docker-compose up -d
Creating network "docker-nginx-task_default" with the default driver
Creating custom-nginx ... done
```

```
root@learninglinux:/home/docker-nginx-task# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
7d8dd0ca03ae	docker-nginx-task_nginx-server	"/docker-entrypoint..."	14 seconds ago	Up 12 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	custom-nginx

```
root@learninglinux:/home/docker-nginx-task# docker ps
```

```
root@learninglinux:/home/docker-nginx-task# curl localhost:8080
<html>
<head>
<title>Nginx Docker</title>
</head>

<body>
<h1>Nginx running from custom docker image</h1>
<h2>Bind mount location: /var/opt/nginx</h2>
</body>

</html>
```

← → ↻ 📄 ⚠ Not secure 192.168.0.45:8080

Nginx running from custom docker image

Bind mount location: /var/opt/nginx